



PS02 : DRIFT-SENSE: AI-POWERED NAVIGATION-ERROR RECOVERY FOR WAFER INSPECTION TOOLS

SOLUTION :

Drift-Sense locates a high-resolution semiconductor reference pattern inside a larger, noisier search image and predicts its centre coordinates using both classical computer vision and deep learning.

Team Details

Team Name:

The Nanoscale Collective

SR.NO	ROLE	NAME	ACADEMIC YEAR
1	Team Leader	Pratyush Khodade	3 rd Year
2	Member 1	Prasad V. Sonar	3 rd Year
3	Member 2	Sunny Nigam	2 nd Year
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Problem Understanding and Navigation-Error Context

The Problem :

A semiconductor inspection system must determine where a small reference region exists inside a much larger search image.

Input Specification :

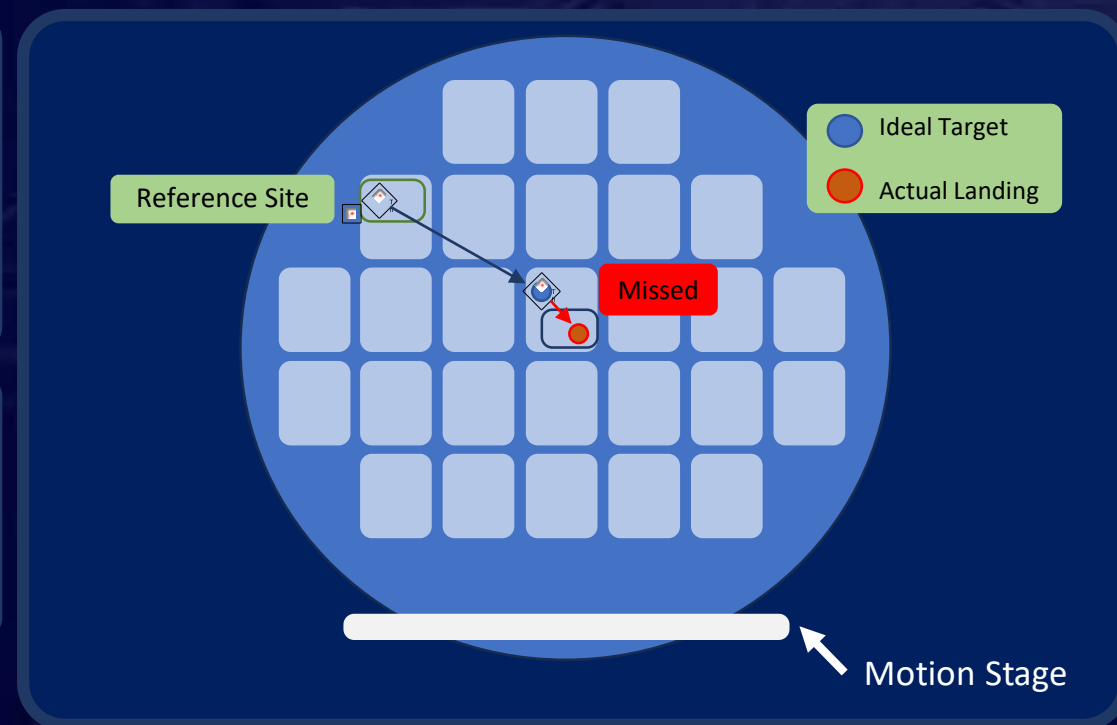
- | Reference Image : 1000 x 1000 px
- | Search Image : 1000 x 1000 px
- | Search Image $\approx$ 10 $\times$ larger physical field of view
- | Reference Image = shrunk 1/10th inside the search image

Why is this Difficult ?

- | DRAM and Fin-FET Layouts are highly repetitive.
- | A search region can contain many visually similar candidates:
:: **Correct pattern $\neq$ only visually similar pattern**
- | Therefore, a model can confidently select the **wrong repeated tile**.

Core Challenge :

Navigation error is not simply a pixel-level positioning problem -- it can become a fundamental pattern-identification problem.



Proposed End-to-End Workflow

DriftSense Localization Workflow



DRAM/Fin-FET Synthetic Data Design

Why Synthetic Data ?

The problem requires realistic semiconductor structures while avoiding dependence on confidential fab data.

DRAM Style Structures :

- | Periodic horizontal **word lines**
- | Vertical **bit lines**
- | Contact/via structures at intersections
- | Dense repeated geometry

Fin-FET Style Structures :

- | Dense parallel **fin lines**
- | Horizontal gate structures
- | Repetitive nanoscale geometry

Important innovation: Gap-collapse defect injection

- | Adjacent DRAM lines / Fin-FET fins can be partially fused to deliberately break perfect periodicity. *collapse_prob = 0.12*
- | This reduced the ambiguity between repeated tiles and became the single largest accuracy improvement.
- | Classical accuracy increased from approximately 1–5% → 17.5–24% @5px, a 5–8× improvement from defect injection alone.

Canonical Datasets :

- | 800 reference/search pairs
- | Defect injection
- | Full degradation set
- | Variable scale: 9×–11×
- | Seed = 2026

Noise, Blur, Scale & Rotation Modelling with Citations



References & Citations

Sensor Noise — Poisson-Gaussian Signal-Dependent Model

DOI : <https://doi.org/10.1109/TIP.2008.2001399>

SEM Charging Artifacts

DOI : <https://doi.org/10.1063/1.336493>

DRAM / Fin-FET Pitch and Structure Scaling

DOI : https://irds.ieee.org/images/files/pdf/2023/2023IRDS_MET.pdf

Wafer Stage Drift and Overlay Error

DOI : <https://doi.org/10.1117/12.2045715>

Key physical model :

| For Poisson-Gaussian noise: $\sigma(x) = \sqrt{a \cdot x + b}$

This models noise as dependent on the underlying signal instead of using only constant additive Gaussian noise.

Important scope limitation :

- Astigmatism and spot-size blur are **not separately modelled as distinct physical effects**; they are represented through the general blur model.
- This is explicitly documented as a scope decision rather than being claimed as full physical modelling.

Localization Method and Decision Rule

Target Localization Methods and Evaluation



Decision Rule in Classical CV Approach :

Select the candidate with the highest-quality match score (within 90% of the global best), then use spatial proximity as the tiebreak.

Decision Rule in DL Approach :

For prediction (x_p, y_p) and ground truth (x_g, y_g) :
Localization error $e = \sqrt{(x_p - x_g)^2 + (y_p - y_g)^2}$

A prediction is counted as correct when:
 $e \leq \text{chosen tolerance}$

Implementation and Execution Commands

Deep Learning Approach for Localization



Classical Computer Vision Approach



Execution Commands for Classical CV Pipeline :

```
python evaluate.py \
  --manifest data/train_canonical/manifest.csv \
  --csv-out results/predictions.csv
```

Execution Commands for DL Pipeline :

```
python evaluate_dl.py \
  --manifest data/train_canonical/manifest.csv \
  --csv-out results/predictions_dl.csv
```

Execution Commands for Merging the Results :

```
python merge_results.py \
  --manifest data/train_canonical/manifest.csv \
  --classical results/predictions.csv \
  --dl results/predictions_dl.csv \
  --out results/full_results.csv
```

Confidence Analysis :

```
python pr_curve.py
```

Experiment Setup and Test Diversity :

Variables tested

Scale

Tested at 9x, 10x, and 11x.

Rotation

Approximately $\pm 3-4^\circ$ of rotation.

Noise

Tested across low, medium, and high terciles.

Defects

Includes gap-collapse injection and process variation or degradation.

Algorithm Testing and Evaluation

Algorithm Testing and Evaluation

Cross-dataset Test

Four Combinations

Real-chip Test

Grayscale Variant

RGB-aware Variant

Clean-environment Test

~30 Sample Pairs

Evaluation Scripts Verified

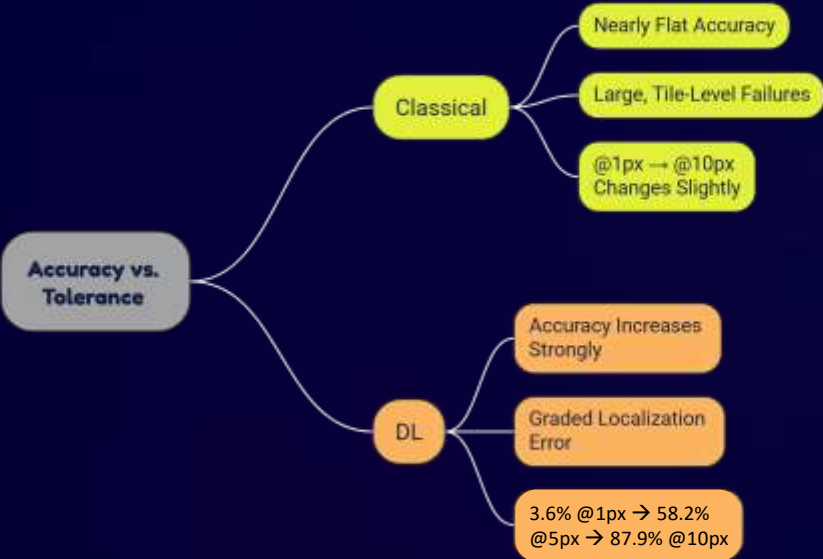
DL Dry-run Training

Threshold-wise Accuracy, Error Statistics and Run Time :

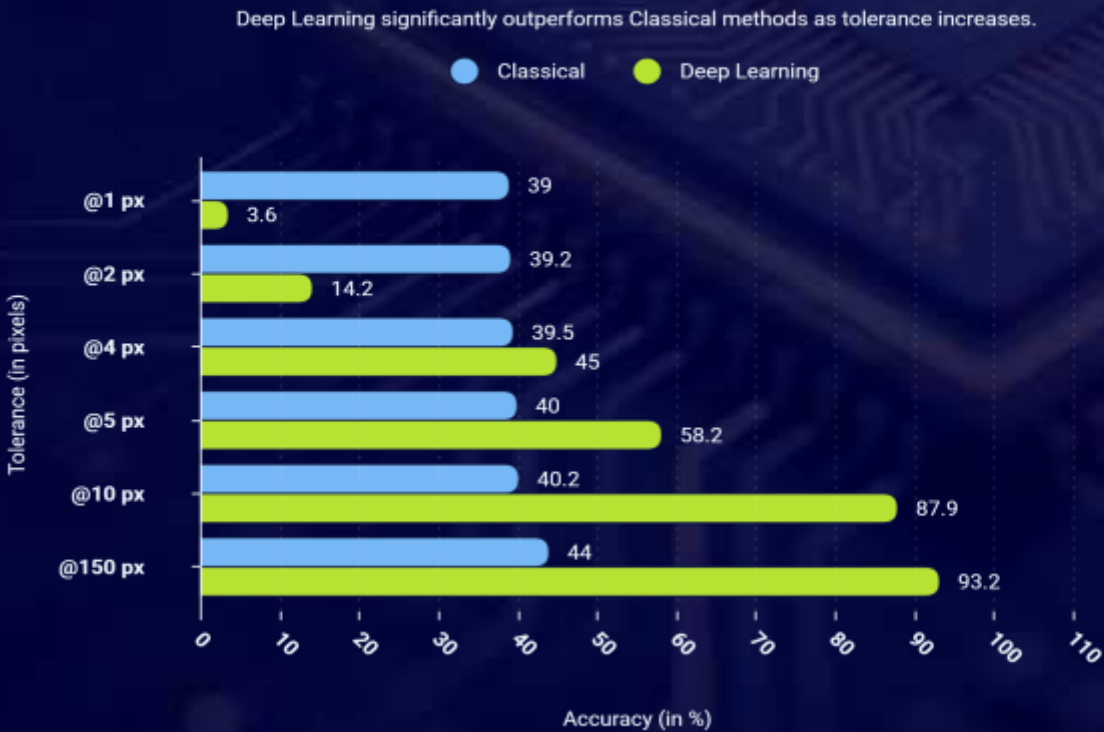
Main Conclusion :

DL provides both higher localization accuracy and lower inference time on the canonical defect-injected, variable-scale dataset.

Accuracy vs. Tolerance in Classical and DL Models

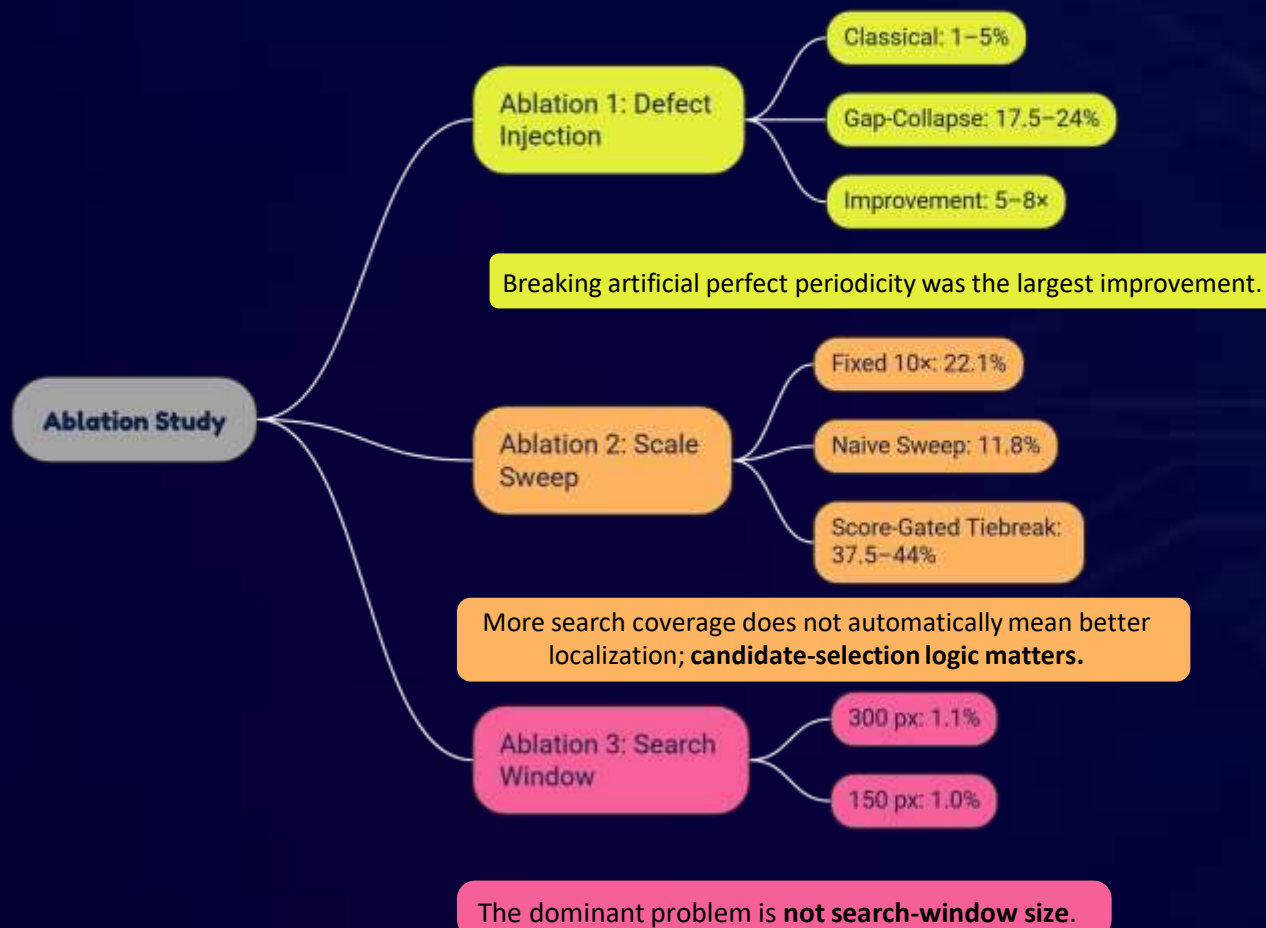


Comparison of Classical and Deep Learning Performance by Tolerance

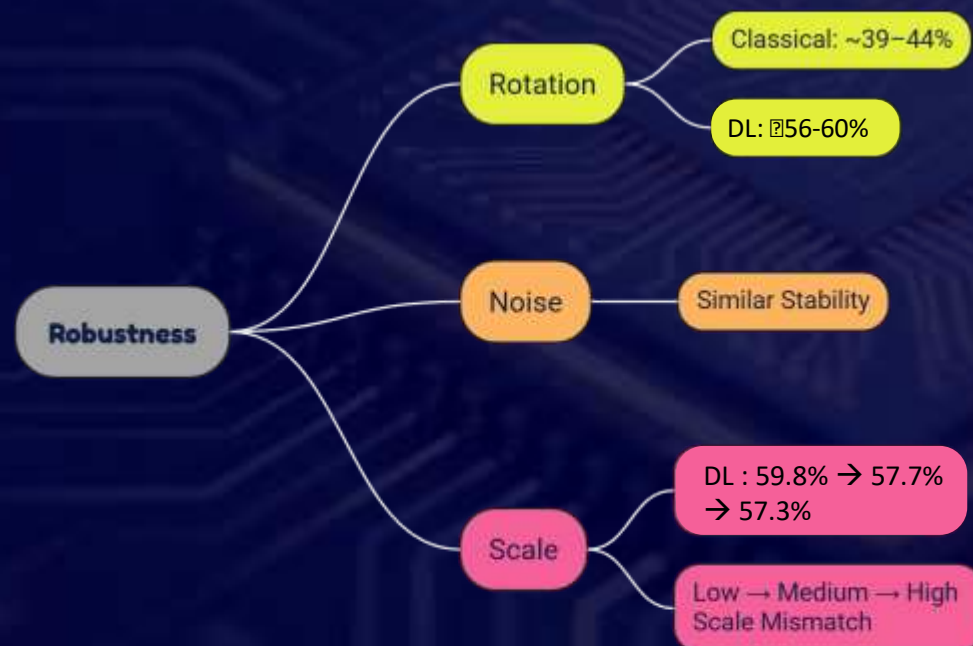


Robustness Comparison and Baseline/Ablation :

Ablation Study Results and Conclusions



Robustness of Deep Learning Models



Failure Case, Limitation and Learning :

Major Failure Modes

Failure Mode	 Dense Periodic Ambiguity	 Scale-Sweep Tiebreak Dilution	 DL Confidence Miscalibration	 Cross-Dataset Domain Shift	 Classical vs DL Failure Behaviour	 Real-Chip Generalization Gap
Status	Mitigated	Fixed	Known limitation	Diagnosed, not solved	N/A	Preliminary
Learning	The problem is feature ambiguity, not simply insufficient search area.	Candidate quality must be considered before spatial proximity.	Accuracy and confidence are not automatically correlated.	Generalization depends heavily on dataset diversity.	Classical is binary failure; DL is gradual failure.	Not a statistically powered real-chip evaluation.

Conclusion, Next Steps, GitHub & Demo Video Link :

Research Improvement Strategies

Strategy	 1. Hybrid Classical + DL Pipeline	 2. DL Confidence Calibration	 3. More Diverse Training Data	 4. Real-Chip Validation	 5. RGB / Optical Microscope Generalization	 6. Better Physical Degradation Models
Description	Use classical CV for coarse localization, then apply DL to a cropped region for fine localization.	Perform a proper calibration study, e.g. temperature scaling on a held-out dataset.	Train across multiple independently generated datasets and real SEM sources.	Expand from one real chip image to a statistically meaningful real-image benchmark.	Address the bonus criterion using RGB optical-microscope images.	Separate: Astigmatism, Spot-size blur, Other SEM-specific imaging effects

Conclusion :

What Drift-Sense demonstrated :

- | Synthetic SEM-like data can be made substantially more useful by modelling realistic degradation and breaking perfect periodicity.
- | Dense periodicity is the dominant localization challenge.
- | Gap-collapse defect injection produced the largest improvement.
- | Score-gated candidate selection solved the scale-sweep regression.

GitHub Repository



<https://github.com/20251414005-ui/The-Nanoscale-Collective-Drift-Sense>



Prototype / Simulation Video

https://youtu.be/J4DrD7h_Oh4?si=urxMeUpCNjxsf9Y